ZUL IKRAM MUSADDIK RAYAT

Lalbagh Road, Dhaka • +880 1580-863471 • rayathossain49@gmail.com • linkedin.com/in/zim-rayat
• github.com/devrayat000 • rayat.dev • v2.rayat.dev • ORCID: 0009-0003-3661-7731

Summary

M.Sc. in Data Science student at BUET CSE and Mechanical Engineering graduate from BUET intentionally transitioning into Software Engineering, bringing strong mathematical foundations, systems modeling, and production experience. Completed a 5-month software engineering internship at RubizCode IT developing backend APIs and AI systems in NestJS, FastAPI, and Redis for 100,000+ users. Comfortable across the stack, from distributed backend pipelines and on-device runtimes to web platforms. Evening/flexible graduate schedule allows 100% full-time professional availability for software engineering roles.

Education

M.Sc. in Data Science **2026 – Present**
Department of Computer Science and Engineering, BUET
- Evening / flexible program • Full-time professional software engineering availability
- Focus: Machine Learning Systems, Advanced Algorithms, Distributed Systems

B.Sc. in Mechanical Engineering **June 2026**
Bangladesh University of Engineering and Technology (BUET)
- CGPA: 3.10 / 4.00 • Former Class Representative

Higher Secondary Certificate (HSC) **2020**
Notre Dame College, Dhaka
- GPA: 5.00 / 5.00

Software Engineering Experience

Software Engineering Intern - Backend & AI Systems **May 2025 – Sept 2025**
RubizCode IT
- Built and scaled **Taalaash**, an academic search engine serving **100k+ users**, **30k+ active students**, and **2k+ daily queries** with **99.9% uptime**.
- Engineered an OCR & text-extraction pipeline using **FastAPI** and fine-tuned **Tesseract OCR**, indexing **20k+ textbook pages** with **98% character recognition accuracy** for Bangla script.
- Implemented semantic search, full-text indexing, and vector reranking using **PostgreSQL**, **Algolia**, and **Pinecone**, lifting search relevance by **35%** and cutting page indexing time by **80%**.
- Developed **TopperOn Doubts AI**, architecting an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline with **FastAPI**, **LangChain**, and **OpenAI API**, reducing query latency by **35%** and boosting context-relevance scores by **2 points**.
- Architected modular backend **microservices** using **NestJS**, **Prisma**, and **PostgreSQL**, incorporating **Redis** caching and multi-layer CDN strategies to improve data throughput and system efficiency by up to **95%**.
- Implemented real-time interactive exam and polling engines using **WebSockets** and asynchronous message queues to handle concurrent live student participation.
- Containerized microservices using **Docker** and established deployment workflows on **GCP** managed behind **Traefik** reverse proxy and **Cloudflare** for automated routing, SSL, and load balancing.

Extracurricular Activities

President **Mar 2025 – Present**
BUET Animata - Animation & VFX Club
- Served as Founding Assistant General Secretary; organized workshops on basic animation and 3D modeling.
- Designed and animated the official club logo and visual assets.

Web Master **Oct 2022 – Oct 2024**
Team Interplanetar - BUET Mars Rover Team
- Developed and maintained the home website of Team Interplanetar containing detailed descriptions of its products, findings and achievements.
- Forged a new certifications system in the website to provide certifications on engineering aspects of mars rovers.

Skills

Languages: JavaScript, TypeScript, Python, Go, Rust, C/C++, C#, SQL
Frontend & Mobile: React, Next.js, React Native, Expo, Redux, Svelte, Tailwind CSS, Framer Motion, React-Query, Jotai, Zustand, Vite
Backend: Node.js, Nest.js, Express.js, Hono, FastAPI, Django, .NET Core, GraphQL, REST
AI & Machine Learning: RAG (Retrieval-Augmented Generation), LangChain, Pinecone, OpenAI API, Tesseract OCR, TensorFlow, Scikit-Learn, NumPy, Pandas
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis (Streams, Pub/Sub), Cloud Firestore, Pinecone (Vector DB)
DevOps & Infrastructure: Docker, Docker Swarm, Traefik, Cloudflare, GitHub Actions, AWS, Google Cloud Platform (GCP), Vercel, Heroku
Tools & Middleware: RabbitMQ, Celery, FFmpeg, Git, Linux, Webhooks, gRPC, LaTeX

Notable Projects

Kabbo - Distributed Polyglot Microservices Platform **Personal**
Go, TypeScript (Hono), Python (FastAPI), PostgreSQL, Redis Streams, Traefik v3, Docker

- Engineered a multi-tenant B2B microservices platform with **Traefik v3** API gateway performing zero-hop **ES256 JWT** validation, rate limiting, and CORS enforcement.
- Implemented per-domain **PostgreSQL** schema isolation with dedicated role credentials; orchestrated asynchronous domain events via **Redis Streams** and BullMQ across 13 containerized services.
- Source code: devrayat000/distributed-system

ScabieSense - AI Health Screening App **BUET IAT Research**
React Native, Expo, TypeScript, Deep Learning

- Built a cross-platform mobile app for early scabies detection using a custom deep learning model, developed for BUET's Cyber Physics Intelligence Lab and field-tested on university dorm students.
- Source code: devrayat000/scabies-net

Video Transcoding Pipeline **Personal**
Go, Redis Streams, FFmpeg, React/Vite

- Designed a multi-service video transcoding system with **Redis Streams** consumer groups for durable, at-least-once job distribution across horizontally-scalable workers.
- Built the worker backend in **Go** for high-throughput **FFmpeg** processing, paired with a real-time **React/Vite** dashboard for monitoring job status and failure modes.
- Source code: devrayat000/video-processing

Retina AI - Distributed Document Ingestion Pipeline **Freelance**
TypeScript, Python, Redis (BullMQ), S3, Docker, Gemini, AI SDK

- Designed a multi-service, queue-based pipeline: a Node.js API enqueues jobs to Redis (BullMQ); horizontally-scalable, independently-deployed Python workers pull jobs, extract and OCR documents (PyMuPDF, Tesseract), and hand off structured output to a downstream embedding queue via S3-compatible object storage.
- Decoupled extraction and embedding into stateless services that scale independently (configurable worker replica count) and communicate only through queues and object storage, with Docker Compose and PM2 deployment paths.
- Source code *available on request*

Focus Stream - On-Device AI Focus Journal **Personal**
Rust, Tauri 2, Llama 3.2 (on-device), SQLite

- Built a fully offline desktop app running a local Llama 3.2 model (via mistral.rs) for on-device AI-generated daily summaries, with no cloud dependency or telemetry.
- Designed a compile-time-selectable inference backend (in-process linking vs. a dynamically-loaded DLL) to support fast dev iteration and a slim production binary across x64/arm64 builds; published to the Microsoft Store.
- Source code *available on request*

Assistive BdSL Gesture Generation on a Robotic Hand **Undergraduate Project**
Python, MediaPipe, TensorFlow, Arduino, Flex Sensor Servo Motor

- Built a real-time speech-to-sign pipeline (ASR → landmark-based MLP classifier → servo control), achieving 91.11% test accuracy across 38 gesture classes on a 3D-printed robotic hand prototype.

Rover Camera Feed **Team Interplanetar - BUET Mars Rover Team**
Python, OpenCV, Flask, WebSockets, WebRTC, Live-Stream

- Built a high-performance multi-camera streaming system for robotics teleoperation, using low-latency WebSocket transmission and OpenCV-based video processing across simultaneous camera feeds.
- Source code: devrayat000/rover_camera_feed

Problem Solving, Certifications & Publications

- **Algorithmic Problem Solving:** Active C++ practice on **LeetCode** and **NeetCode** (devrayat000/neetcode-submissions).
- **Published IEEE Papers (3):** Author/co-author on 3 peer-reviewed IEEE publications: Hard-Constraint Navier-Stokes PINNs, Spatiotemporal PM2.5 Attention Forecasting, and 6-DOF Robotic Hand BdSL Kinematics.
- **Foundational CS Certifications:** CS50x (Harvard) • Algorithms Specialization (Stanford) • Algorithms, Part I & II (Princeton).
- **Honors & Awards: Finalist**, Build4Democracy Hackathon (UNDP, 2025) • **Finalist**, Innovation Founder (RISE UIHP, 2024) • **1st Place**, Speak Out For Engineers (IMechE BUET Student Chapter, 2023) • **Technical Scholarship**, BUET (2022, 2023).

Reference

Dr. Mohammad Mamun
Professor
Dept. of Mechanical Engineering, BUET
Email: dmmamun@me.buet.ac.bd

Dr. Mohammad Jane Alam Khan
Associate Professor
Dept. of Mechanical Engineering, BUET
Email: ronin@me.buet.ac.bd